Supplementary Material for "Reinforcement Learning for Flexible Job Shop Scheduling: A Systematic Review of End-to-End and Steering Policies"

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Abstract

This Supplementary Material reports the search terms, screening model settings, calibration and validation results, and complete map coding scheme. It also reports the deterministic core-selection procedure, record flow, full-text coding details, and the 99 core studies. It accompanies the concise method reported in the paper.

1. Search strategy and query terms

The OpenAlex search [1] implemented the eligibility vocabulary directly. We joined 37 reinforcement-learning method terms with Boolean `OR`, and did the same for 25 flexible job-shop problem terms. We combined the two facets with Boolean `AND` over `title_and_abstract` and restricted the document type to `article` or `conference-paper`. Table 1 gives the exact frozen lists.

Table 1: Closed lists of 62 search terms, comprising 37 method terms and 25 problem terms. The lists were combined through Boolean `AND` over `title_and_abstract` in OpenAlex on 11 October 2026.

Facet and group	Exact terms queried in OpenAlex
Method: RL core (1)	`reinforcement learning`
Method: Algorithms (27)	`q learning`; `q network`; `q networks`; `deep q network`; `deep q networks`; `dqn`; `dqns`; `sarsa`; `reinforce`; `policy gradient`; `policy gradients`; `actor critic`; `a2c`; `a3c`; `proximal policy optimization`; `proximal policy optimisation`; `ppo`; `trust region policy optimization`; `trust region policy optimisation`; `trpo`; `deep deterministic policy gradient`; `ddpg`; `twin delayed ddpg`; `td3`; `soft actor critic`; `sac`; `temporal difference learning`
Method: Adjacent (9)	`policy`; `policies`; `markov decision process`; `mdp`; `mdps`; `markov`; `markovian`; `decision process`; `decision processes`
Problem: Names (10)	`flexible job shop`; `flexible jobshop`; `flexible job shops`; `flexible jobshops`; `flexible assembly job shop`; `flexible assembly jobshop`; `flexible assembly job shops`; `flexible assembly jobshops`; `flexible workshop`; `flexible workshops`
Problem: Abbreviations (15)	`fjs`; `fjsp`; `fjsps`; `fjssp`; `fjssps`; `dfjsp`; `dfjsps`; `dfjssp`; `dfjssps`; `mofjsp`; `mofjsps`; `fjspt`; `fjspts`; `fajsp`; `fajsps`

Before freezing the lists, we compared OpenAlex results for spaced, hyphenated, singular, and plural forms. OpenAlex treated hyphens inside multiword phrases as spaces but did not fold plurals. An isolated hyphenated token was translated into component words rather than an exact phrase. We therefore used spaced phrases, listed plurals and British spellings explicitly, and retained abbreviations that added eligible studies. Candidate terms outside the initial lists were tested for marginal yield. Only `q network` and `fjs` added eligible records not already found by another listed term.

The one official search, run on 11 October 2026, returned 664 works. OpenAlex supplied no abstract for 288 of them. We recovered 191 of these abstracts from Scopus, 71 from the Springer Nature Meta API, 16 from Semantic Scholar, and six from publisher pages. Four records still lacked an abstract. We removed two duplicates under the pre-specified title and first-author rule. We dropped three records with neither an abstract nor a DOI under the protocol's search rules, leaving 659 records to screen.

2. Screening panel, calibration, and validation

The screening panel comprised three open-weight language models from distinct developers, with the pinned settings in Table 2. Each record was evaluated in an independent request containing the eligibility criteria and the record's title, abstract, venue, year, and document type. The request required `include`, `exclude`, or `uncertain`, a reason beginning with the deciding criterion, and a supporting quotation. Outputs without a criterion label were requested again.

Table 2: Open-weight language models and pinned inference settings used for title-and-abstract screening and map coding.

Setting	Qwen3.8-27B	Gemma 4 26B A4B	Muse Glimmer 30B
Developer	Alibaba	Google	Meta
Hugging Face repository	`Qwen/Qwen3.8-27B`	`google/gemma-4-26B-A4B-it`	`meta-models/Muse-Glimmer-30B`
Pinned weight revision	`1d4bf0f2...c60c0`	`4d7ae498...76d52`	`a4e59da5...c44b68`
Weight precision	Native (unquantized)	Native (unquantized)	Native (unquantized)
Reasoning mode	Enabled (effort: medium)	Enabled (thinking)	Enabled (strength: medium)
Temperature / Top-p / Top-k	1.0 / 0.95 / 20	1.0 / 0.95 / 64	1.0 / 0.95 / 64
Context / Max output tokens	49,152 / 32,768	49,152 / 32,768	49,152 / 32,768
Serving engine and hardware	vLLM 0.31.0, 1× H100 80 GB	vLLM 0.31.0, 1× H100 80 GB	vLLM 0.31.0, 1× H100 80 GB

The vote was asymmetric to limit false exclusions. A record was included whenever at least two models voted `include`. Otherwise, any `exclude` vote removed it as excluded. A record with no exclusion and fewer than two inclusions was removed as uncertain. This settlement rule was adopted on 13 October 2026 after the vote totals were known but before any disagreement was examined.

For calibration, the three models screened the first 25 records, and one author (O.H.H.) screened the same records blind to their decisions. The criteria would be clarified once if any model agreed with the author on fewer than 20 records. Qwen and Gemma each agreed on 25 of 25 records (weighted Cohen's $\kappa = 1.000$), whereas Glimmer agreed on 23 of 25 ($\kappa = 0.648$). Its two differences were uncertain votes on records the author included. The panel vote matched the author on all 25 records, comprising 24 inclusions and one exclusion. Thus, no clarification was triggered.

Across all 659 records, Qwen included 537, left 15 uncertain, and excluded 107. Gemma included 543, left 21 uncertain, and excluded 95. Glimmer included 487, left 54 uncertain, and excluded 118. Pairwise weighted κ ranged from 0.791 to 0.896. The consensus vote included 521 records, unanimously excluded 93, and produced 45 disagreements. The settlement rule included 11 disagreements and removed 34, split evenly between excluded and uncertain.

For validation, we drew a seeded stratified sample outside the calibration batch. It comprised 60 of the 126 removals and 40 of the 508 inclusions decided by the vote or the settlement rule. The same author (O.H.H.) screened these 100 records blind to the models and initially included 15 removals, marked two removals uncertain, and excluded one inclusion. A second author (M.A.) then independently screened all 18 disagreements blind to both the models and O.H.H.'s decisions. The second reading follows Kitchenham and Charters, who recommend that every disagreement be resolved and that a supervisor review a single screener's decisions [2]. The adjudicator included one of the 17 disputed removals, excluded the other 16, and confirmed the disputed inclusion. After adjudication, the validation sample contained one wrong removal among 60 (1.7%, one-sided 95% upper bound 7.7%). It contained no wrong inclusion among 40 (upper bound 7.2%). Its weighted κ was 0.979. Carrying the one adjudicated record forward produced 533 inclusions, 110 exclusions, and 16 uncertain records.

3. Complete map coding scheme

We coded the 533 included records on 15 axes from their titles and abstracts. Table 3 gives every substantive code, the models' agreement, and whether the axis could steer core coverage. The map answers Question 1 and supplies the sampling frame for the core.

Table 3: The 15 map axes coded from titles and abstracts across the 533 included studies. The table adds inter-model agreement across the 532 model-coded records and the core-coverage decision.

Axis	RQ	Type	Substantive codes (excluding `not-stated`, `other`, and `not-applicable`)	Mean κ	Unanimous (%)	Steers core
`policy_role`	1.1	Single	`end-to-end`; `steering-rules`; `steering-heuristic`	0.817	74.6	Yes
`coordination`	1.1	List	`single-agent`; `multi-agent`; `hierarchical`	0.890	92.3	Yes
`scheduling_action`	1.1, 2.1	Single	`operation`; `machine`; `operation-machine-pair`; `whole-schedule`; `schedule-change`	0.565	85.3	No
`learner_depth`	1.2	Single	`deep`; `non-neural`; `tabular`	0.978	97.9	Yes
`deep_architecture`	2.2	List	`fully-connected`; `convolutional`; `recurrent`; `graph-neural-network`; `attention-transformer`; `pointer-network`	0.897	87.6	Yes
`training_algorithm_family`	1.2	List	`value-based`; `policy-gradient`; `actor-critic`	0.986	97.7	Yes
`hybridization`	1.2	Single	`none`; `metaheuristic`; `exact`; `other-learner`	0.931	95.1	Yes
`state_content`	2.1	List	`operations`; `jobs`; `machines`; `shop-aggregate`; `events`; `other-resources`; `search-state`	0.665	69.7	Yes
`state_encoding`	2.2	List	`discrete-states`; `feature-vector`; `sequence`; `set`; `matrix`; `disjunctive-graph`; `heterogeneous-graph`; `other-graph`; `text`	0.743	82.1	Yes
`dynamic_events`	1.3	List	`none`; `job-arrivals`; `machine-breakdowns`; `uncertain-processing-times`	0.944	87.4	Yes
`objective`	1.3	List	`makespan`; `tardiness`; `flow-time`; `workload`; `utilization`; `energy`; `cost`; `robustness`	0.914	87.8	Yes
`constraints`	1.3	List	`none`; `transport`; `setup-times`; `workers`; `energy`; `distributed`	0.887	76.1	Yes
`instances`	3.1	List	`public-benchmark`; `generated`; `industrial`	0.821	82.9	Yes
`domain`	1.3	List	`manufacturing`; `computing-networks`	0.882	99.2	Yes
`readiness_claim`	5	Single	`none`; `real-data`; `twin-or-hil`; `testbed-or-shadow`; `live-system`	0.644[a]	92.7	Step 1 only

[a]Multiclass Cohen's κ across all five readiness levels; `readiness_claim` enters Step 1 after author confirmation rather than steering Step 2.

The scheme was frozen on 14 October 2026 after a pilot on the 24 included records of the calibration batch. Models used `not-stated` when an abstract did not provide an axis, except that `hybridization`, `domain`, and `readiness_claim` had explicit

defaults. The three models coded 532 records before screening was released. Record 74, admitted through validation adjudication, was coded by M.A. under the same scheme and excluded from inter-model agreement.

Five deterministic rules made dependent axes consistent before and after voting:

1. A steering role sets `scheduling_action` to `not-applicable`.
2. An unstated role sets `scheduling_action` to `not-stated`.
3. A tabular or non-neural learner sets `deep_architecture` to `not-applicable`.
4. An unstated learner sets `deep_architecture` to `not-stated`.
5. A tabular learner sets `state_encoding` to `discrete-states`.

Each single-code axis was resolved by majority vote, and each list code was voted independently. A cell without a majority became `not-stated`. Readiness instead took the highest level reached by at least two models. Agreement for Step 2 was computed by treating every substantive code as present or absent. Pairwise Cohen's κ was averaged across model pairs and then across the axis's substantive categories. Thirteen of 14 candidate axes reached 0.60; `scheduling_action` did not. A seeded 25-record author audit covered `policy_role`, `learner_depth`, and `readiness_claim`. Agreement with the map was 18 of 25 ($\kappa = 0.615$), 25 of 25 ($\kappa = 1.000$), and 25 of 25, respectively.

4. Core selection and full-text collection

We selected the core through purposive maximum-variation sampling [3]. Of the 533 map records, 50 had no stated learner and remained in the map only. The other 483 were candidates: 156 end-to-end, 180 steering, and 147 role-not-stated by their abstracts, including 388 deep and 95 tabular learners. No candidate was merged with another under the DOI, OpenAlex identifier, or shared-title and matching-first-author rule. At every step, non-deep learners were capped at 10 core places.

The deterministic selection filled 100 places in three sequential steps:

1. Step 1 reserved a place for every candidate whose abstract-level readiness claim an author confirmed at L3–L5. Among 41 candidate claims above L1, 36 were confirmed at L2, four at L3, and one was not confirmed. Only the four L3 studies entered at this step.
2. Step 2 required at least three studies in every substantive category of the 13 reliable coverage axes, or all candidates when fewer than three existed. Starting from Step 1, a greedy rule added the candidate satisfying the most unmet category minima. Ties were resolved by venue tier, within-year citation standing, and record number. This step added 36 studies, including four tabular learners, and met every attainable category minimum.
3. Step 3 filled the remaining 60 places to reach fixed abstract-role strata of 35 end-to-end, 35 steering, and 30 role-not-stated studies. It prioritized Venue Band 1, defined as SJR Q1 or ICORE A*. It then allocated each stratum's places across publication years by largest remainders and selected the highest within-year citation standing. The step added 21 end-to-end, 22 steering, and 17 role-not-stated studies, all from Band 1. Four additional non-deep candidates, records 376, 86, 463, and 384, were bypassed under the cap. Deep candidates 135, 41, 575, and 106 from the same stratum, band, and year replaced them.

The selection software was written and tested after map coding but before any study was selected, and the first run was locked on 19 October 2026. The locked core contained 90 deep and 10 tabular learners, with 91 studies from Band 1. Reversing record-number order in the final tie-break changed eight of the 100 studies. They were four at Step 2 and four at Step 3, all published in 2025 or 2026. We sought all 100 full texts, retrieving and verifying 99. Record 18 was the exception: its proceedings volume was unavailable through the institutional subscriptions. We did not replace it, because the selection was locked before retrieval.

5. Record flow

Table 4 gives the generated flow counts, including Step 3 by abstract-role stratum and venue band. Figure 1 shows the same flow together with map and full-text role resolution.

6. Full-text coding and synthesis details

Part B is the full-text coding scheme. It was piloted on five studies and frozen on 21 October 2026. The scheme codes the 15 map axes again, 54 further structured questions serving Questions 1 through 5, and the seven quality items. This gives 69 structured questions and seven quality items. A substantive answer requires one to three verbatim quotations of at most 50 words with exact page numbers. A paper silent after searching is coded `not-stated` or `not-reported`.

Table 4: Record flow from database identification through title-and-abstract screening, core selection, and full-text collection.

Stage	Records
Identified, databases and registers	664
Duplicates removed	2
Removed for other reasons	3
Screened on title and abstract	659
Excluded at screening	110
Uncertain at screening	16
Included	533
Not yet screened	0
Candidates for the core: learner stated, each study once	483
The core, step 1: use beyond simulation	4
The core, step 2: coverage	36
The core, step 3: the places left	60
The core, step 3: end-to-end, band 1	21
The core, step 3: end-to-end, band 2	0
The core, step 3: steering, band 1	22
The core, step 3: steering, band 2	0
The core, step 3: role-not-stated, band 1	17
The core, step 3: role-not-stated, band 2	0
The core, step 3: non-deep studies left out by the cap	4
Selected for full-text synthesis: the core	100
Full texts sought	100
Full texts not retrieved	1
Full texts coded	99

The coding model was `Qwen3.8-27B` at medium reasoning effort and BF16 precision, served through vLLM 0.31.0 on one NVIDIA B200. Eight papers were processed concurrently, each call allowed up to 16,384 output tokens, and each call received a deterministic seed. In the custom agent, a lead agent delegated sections to section agents whose tools read pages and wrote one answer block at a time. The deterministic checker refused blocks with invalid codes, inconsistent cross-question answers, mismatched item counts, or quotations not found verbatim on the cited page. All 99 papers were coded, and every completed response passed the checker again. The gathered dataset contains 7,524 question answers and 839 item blocks, comprising 368 baselines, 315 claims, and 156 transfer assessments.

The judge was `Muse Glimmer 30B` at medium reasoning strength, also served through vLLM on one B200. Each unit contained all blocks of one question with the paper, or a paper's unquoted answers together. A paper too long for the judge's context was judged through its cited and neighboring pages, or in sequential page windows. The judge evaluated 8,354 blocks: 6,750 supported, 968 partly supported, 605 not supported, and 31 not judgeable. Nine blocks received no valid verdict. The verdicts are published beside the coding.

A blind check of six questions by one author (O.H.H.) was added on 24 October 2026. These questions were the policy role, the training algorithm, the readiness level, the number of training runs, whether every baseline was rerun, and whether a transfer assessment was reported. The author coded them for the 20 studies of the seeded fifth from their full texts, blind to every model's answer and verdict. A second coder on a random sample is a check that Kitchenham and Charters recommend when one coder extracts the data [2]. Table 5 gives the agreement between the author and the custom agent on each question. On readiness, all 11 disagreements were the author's L2 against the custom agent's L1, on records 68, 90, 130, 132, 133, 157, 226, 237, 290, 383, and 447. No disagreement involved a level above L2. Both coders placed record 527 at L2 and record 588 at L4. One difference in the instructions fits this pattern. The custom agent received the scheme's general rules. These rules say to code what the experiments ran, not what a paper says its method could support. They also say that the introduction and the related work decide nothing. The author's sheet gave the readiness definition without these rules, to be applied as written. On training runs, the author found no count reported in all 8 disagreements. The custom agent gave a single run in 7 of them and 10 runs in the eighth. Across the core, the judge supported fewer than half of the training-run answers. We adopt the custom agent's codes for the synthesis, and report the author's codes in the repository.

During drafting, four studies were flagged for author reading. Two were retained as coded. Records 191 and 530 were corrected from steering a heuristic to end-to-end, together with the dependent scheduling-action, sub-decision, steering-action, and steered-method answers. The gather applies these 10 corrections across two studies to the synthesis. It retains the custom agent's original values for coder-judge agreement and the corrected values in the sensitivity analysis without disputed answers.

The seven quality items cover the shop specification, Markov decision process specification, algorithm specification, and repeatable instances. They also cover baseline execution and time limits, several training runs with the variation of their results, and the alignment of transfer or readiness claims with the experiments the study ran. The custom agent scored each item as met, partly



Figure 1: Detailed record flow from the 664 OpenAlex works through screening to the 533-study map and the 483 core candidates. It continues with the three-step selection of 100 studies and the retrieval and coding of 99 full texts. The coded core is filed into 53 end-to-end and 46 steering studies. This diagram was drawn with Claude Opus 5.5 (Anthropic) from the authors' description of the workflow. The authors checked every stage and count against the review's records.

Table 5: Agreement of the author and the custom agent on the 20 studies of the seeded fifth. In each disagreement, the first code is the author's and the second is the custom agent's.

Question	Agree	κ	Disagreements
Policy role	20	1.00	None
Algorithm	19	0.92	Record 132: `q-learning` against `ddpg-td3`
Readiness	9	0.16	11 records: L2 against L1
Training runs	12	0.43	8 records: `not-reported` against 1 (7 records) or 10 (1 record)
Baselines rerun	17	0.58	3 records: `yes` against `no`
Transfer assessment	16	0.53	3 records: `no` against `yes`; 1 record: `yes` against `no`

met, not met, or not assessable, and the judge read every score against the paper. The quality scores define one sensitivity analysis. This analysis leaves out the 10 studies with the lowest scores, and studies tied at the cut are kept.

The synthesis is descriptive. Question 1 and every field-wide count use the 533-study map. Questions 2 through 5 use the 99 full texts. Each core study is filed by the role its full text gives and then by the family of the algorithm training its learned policy. The final filing contains 53 end-to-end and 46 steering studies, of which 24 choose among dispatching rules and 22 steer a heuristic or metaheuristic. Special answers remain explicit denominator rows. A study with multiple codes counts under each. Repeated items are binned as none, one, two, or three or more. Numeric answers are summarized by the reporting share, median, and range.

Question 3.4 uses five comparability checks for each recurring claim. A study meets each check when it:

1. uses a named benchmark set that another study making the claim also uses;
2. reports the same objective with the same statistic as another study making the claim;
3. reruns every baseline on its own instances and states each time limit;
4. meets the first three conditions with one and the same other study;
5. reports how much its results vary across several training runs.

The checks of instances, baselines and time limits, and training runs were fixed before results were seen. The checks of metrics and of all conditions together were added after the initial results to cover every term of the question. We did not check whether studies used the same baselines, because a baseline's name is free text.

Five sensitivity analyses accompany the synthesis:

- comparing the map's abstract codes with the full-text codes for all 99 core studies;
- leaving out the 1,565 answers and items that the judge disputed and the authors did not correct;
- leaving out the 10 studies whose mean quality score lies below the cut for the lowest fifth, because studies tied at the cut are kept;
- leaving out the 36 studies admitted at Step 2 for category coverage;
- recomputing map proportions on the records coded identically by all three models, without record 74 because no model coded it.

The analysis without disputed answers keeps every study in its role and family cell in the list of core studies, but its role-by-family table counts only the answers the judge did not dispute. The paper gives every core count and share in its evidence section and answers table with this analysis's value in square brackets.

7. The 99 core studies

The generated table below gives the synthesis cell and DOI of each coded core study. It lists the 99 retrieved studies.

Table 6: The 99 core studies, with their synthesis cell and DOI.

Record	First author	Year	Role and family	DOI
191	Yingjie Hou	2025	end-to-end / value-based	https://doi.org/10.1016/j.cie.2025.110990
469	Zi-Qi Zhang	2025	end-to-end / value-based	https://doi.org/10.1016/j.asoc.2025.113474
530	Junpeng Su	2022	end-to-end / value-based	https://doi.org/10.1109/tcyb.2022.3158334
582	Ricardo Magalhaes	2021	end-to-end / value-based	https://doi.org/10.1109/ssci50451.2021.9659849
609	Tianfeng Wang	2020	end-to-end / value-based	https://doi.org/10.1109/access.2020.3022038
8	Xiaoyu Yang	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.swevo.2026.102331
24	Kun Lei	2022	end-to-end / policy-gradient	https://doi.org/10.1016/j.eswa.2022.117796
32	Yijie Wang	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.eswa.2025.127671
57	Runqing Wang	2023	end-to-end / policy-gradient	https://doi.org/10.1109/tnnls.2023.3306421
66	Rongkai Wang	2024	end-to-end / policy-gradient	https://doi.org/10.1109/jiot.2024.3485748
68	Xinning Liu	2025	end-to-end / policy-gradient	https://doi.org/10.1007/s40747-024-01772-x
76	B. A. Han	2021	end-to-end / policy-gradient	https://doi.org/10.2507/ijsimm20-2-co7
78	Kun Lei	2023	end-to-end / policy-gradient	https://doi.org/10.1109/tii.2023.3272661
90	Zi-Qi Zhang	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.cie.2025.111466
94	Zhenran Kuai	2022	end-to-end / policy-gradient	https://doi.org/10.1109/tcomm.2022.3211071
108	Wen Song	2022	end-to-end / policy-gradient	https://doi.org/10.1109/tii.2022.3189725
109	Yiwen Hu	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.asoc.2025.113712
120	Junjun Chen	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.asoc.2026.115296
133	Dailin Huang	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.asoc.2025.113128
157	Kaikai Zhu	2026	end-to-end / policy-gradient	https://doi.org/10.1109/tsmc.2026.3656196
158	Xi Liu	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.cor.2025.107139
160	Erdong Yuan	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.asoc.2025.112787
184	Zhaolei He	2026	end-to-end / policy-gradient	https://doi.org/10.3390/asi9100202
201	Chuanzhao Yu	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.jmsy.2025.05.002

Record	First author	Year	Role and family	DOI
213	Alexandre Jesus	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.cor.2026.107428
216	Zhi Pang	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.asoc.2026.114587
228	Zhi Cao	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.swevo.2026.102390
237	Hang Lin	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.eswa.2026.134276
244	Zunxun Wang	2025	end-to-end / policy-gradient	https://doi.org/10.1109/tnnls.2025.3578368
245	Jinsheng Li	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.rcim.2026.103364
255	Ruyi Wang	2026	end-to-end / policy-gradient	https://doi.org/10.3390/su18073222
258	Lingling Lv	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.cie.2025.111256
259	Yuxiang Shen	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.asoc.2026.114899
262	Guangshuai Ning	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.aei.2026.104725
265	Weiyao Cheng	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.eswa.2025.128142
293	Wenqiang Zhang	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.cor.2025.107155
297	Chuanzhao Yu	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.rcim.2025.103180
300	Zepeng Liu	2026	end-to-end / policy-gradient	https://doi.org/10.3390/machines14091018
315	Xize Liu	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.compeleceng.2026.111371
336	Zipeng Yang	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.engappai.2025.111375
339	Zenghui Yi	2026	end-to-end / policy-gradient	https://doi.org/10.1007/s10845-025-02784-1
350	Yuzhi Zhang	2025	end-to-end / policy-gradient	https://doi.org/10.1016/j.eswa.2025.128411
354	Dailin Huang	2024	end-to-end / policy-gradient	https://doi.org/10.1016/j.cie.2024.110768
383	Nanfeng Ma	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.asoc.2026.115016
409	Xuemei Gan	2025	end-to-end / policy-gradient	https://doi.org/10.1115/1.4070035
437	Fuqing Zhao	2026	end-to-end / policy-gradient	https://doi.org/10.1109/tii.2025.3646962
583	David Heik	2024	end-to-end / policy-gradient	https://doi.org/10.1016/j.jmsy.2024.09.019
588	Jie Chen	2026	end-to-end / policy-gradient	https://doi.org/10.1016/j.rcim.2026.103323
46	Lanjun Wan	2024	end-to-end / actor-critic	https://doi.org/10.1016/j.knosys.2024.111940
77	Wenquan Zhang	2024	end-to-end / actor-critic	https://doi.org/10.1016/j.jmsy.2024.03.012
101	Xuan Jing	2022	end-to-end / actor-critic	https://doi.org/10.1007/s10845-022-02037-5

Record	First author	Year	Role and family	DOI
214	Chupeng Su	2024	end-to-end / other	https://doi.org/10.1016/j.swevo.2024.101605
504	Yuzhi Zhang	2026	end-to-end / not-stated	https://doi.org/10.1016/j.cie.2026.111897
1	Minghai Yuan	2023	steering / value-based	https://doi.org/10.1007/s10845-023-02252-8
21	Yuxin Li	2021	steering / value-based	https://doi.org/10.1016/j.rcim.2021.102283
41	Yu Du	2022	steering / value-based	https://doi.org/10.1109/tnnls.2022.3208942
52	Yuhang Yan	2025	steering / value-based	https://doi.org/10.1109/access.2025.3564433
65	Lei Yue	2024	steering / value-based	https://doi.org/10.1016/j.swevo.2024.101660
72	Shaojun Lu	2024	steering / value-based	https://doi.org/10.1016/j.engappai.2024.108487
92	Rui Li	2023	steering / value-based	https://doi.org/10.1109/tsmc.2023.3305541
103	Xiaoting Dong	2025	steering / value-based	https://doi.org/10.1007/s40747-025-01828-6
106	Huynh Hoang My Dung	2026	steering / value-based	https://doi.org/10.1016/j.asoc.2026.116554
127	Fujie Ren	2026	steering / value-based	https://doi.org/10.1016/j.future.2026.108450
130	Weiyao Cheng	2025	steering / value-based	https://doi.org/10.1016/j.cor.2025.107087
131	Jian Ma	2025	steering / value-based	https://doi.org/10.1016/j.engappai.2025.110447
135	Liang Zheng	2025	steering / value-based	https://doi.org/10.1080/00207543.2025.2497961
136	Yong Lei	2025	steering / value-based	https://doi.org/10.1016/j.asoc.2025.114549
140	Yu Du	2022	steering / value-based	https://doi.org/10.1109/tetci.2022.3145706
161	Xiao Chang	2024	steering / value-based	https://doi.org/10.1080/00207543.2024.2357740
204	Tianen Li	2026	steering / value-based	https://doi.org/10.1016/j.cor.2026.107500
211	Fayong Zhang	2024	steering / value-based	https://doi.org/10.1016/j.cie.2024.109917
226	Lan Yang	2026	steering / value-based	https://doi.org/10.1016/j.orp.2026.100379
249	Xinxin Zhou	2025	steering / value-based	https://doi.org/10.1016/j.swevo.2025.101885
274	Tairan Fu	2026	steering / value-based	https://doi.org/10.1016/j.cie.2026.111903
290	Rensheng Chen	2024	steering / value-based	https://doi.org/10.1016/j.swevo.2024.101658
353	Xi Zhang	2026	steering / value-based	https://doi.org/10.1016/j.cor.2026.107591
393	Zhijie Yang	2025	steering / value-based	https://doi.org/10.1016/j.jmsy.2025.02.002
447	Zhengxin Zhang	2026	steering / value-based	https://doi.org/10.1049/cim2.70077

Record	First author	Year	Role and family	DOI
465	Tianhua Jiang	2026	steering / value-based	https://doi.org/10.1016/j.eswa.2025.131024
477	Tianen Li	2026	steering / value-based	https://doi.org/10.1016/j.eswa.2026.132032
503	Fuqing Zhao	2026	steering / value-based	https://doi.org/10.1109/tsmc.2025.3648993
555	Fuqing Zhao	2025	steering / value-based	https://doi.org/10.1109/tcyb.2025.3538007
565	Zhengpei Zhang	2024	steering / value-based	https://doi.org/10.1016/j.cie.2024.110484
574	Yaping Fu	2025	steering / value-based	https://doi.org/10.1109/tetci.2025.3540422
575	Wenchao Yi	2026	steering / value-based	https://doi.org/10.1016/j.asoc.2026.116088
577	Candice Destouet	2025	steering / value-based	https://doi.org/10.1016/j.cor.2025.107323
606	Erickson Liang	2026	steering / value-based	https://doi.org/10.1016/j.ijpe.2026.109979
12	Lu Zhang	2023	steering / policy-gradient	https://doi.org/10.1016/j.jmsy.2023.09.009
40	Chunyan Duan	2026	steering / policy-gradient	https://doi.org/10.1016/j.cor.2026.107700
54	Kjell van Straaten	2026	steering / policy-gradient	https://doi.org/10.1007/978-3-032-09156-7_13
197	Dongping Qiao	2026	steering / policy-gradient	https://doi.org/10.3390/math14111914
247	Jiaxuan Shi	2025	steering / policy-gradient	https://doi.org/10.1016/j.aei.2025.103195
291	Hao Wang	2025	steering / policy-gradient	https://doi.org/10.1016/j.eswa.2025.126441
357	Weiyuan Wang	2026	steering / policy-gradient	https://doi.org/10.1016/j.eswa.2026.131816
399	Wenbin Gu	2024	steering / policy-gradient	https://doi.org/10.1016/j.cie.2024.110155
572	Zhao Yixun	2024	steering / policy-gradient	https://doi.org/10.1109/iccea62105.2024.10603738
15	Yong Gui	2023	steering / actor-critic	https://doi.org/10.1016/j.cie.2023.109255
132	Xiao Wang	2025	steering / actor-critic	https://doi.org/10.1109/access.2025.3577044
527	Yue Liang	2026	steering / actor-critic	https://doi.org/10.3390/app16178807

8. Software and data

The repository at https://github.com/malnammi-lab/rl-fjsp-review contains the software code used to conduct this review, a supplement with further details, and the bibliography of the 533 studies.

Author contributions

Moayad Alnammi: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing, Project administration. **Osamah H. Hussein:** Conceptualization, Data curation, Investigation, Validation, Writing – review & editing. **Ahmed Azab:** Conceptualization, Methodology, Investigation, Writing – review & editing, Project administration. **Yasser Almoghathawi:** Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude Opus 5.5 (Anthropic) to improve the clarity and readability of text they had written. The authors also used the same tool to draft code under their guidance and according to their specifications. The diagram in Fig. 1 was drawn with the same tool from the authors' specification and description. Throughout these uses, the authors remained actively involved and supervised the outputs. The authors reviewed, edited, verified, and take full responsibility for the content of this work.

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